Chemical Kinetics

(10/9/2020)

A elementary step in a reaction of the form

$A \rightarrow P$

has a rate law of the form:

$$v = -\frac{d[A]}{dt} = \frac{d[P]}{dt} = k[A]$$

where v is the reaction rate, t is time, k is rate constant, and [A] and [P] are concentrations of reactant and product, respectively.

1. What is the order of the above reaction (A→P)? What are the units of k? (10 pts)

2. Derive and write the expression of the integrated rate law of the above reaction (A→P). (20 pts)

3. Derive and write the expression of the half-life ($t_{1/2}$) of the above reaction (A→P). Half-life is the time required for half the initial concentration ($[A]_0$) to react. (20 pts)

4. The decomposition of benzoyl peroxide in diethyl ether, ($(C_6H_5COO)_2 \rightarrow 2C_6H_5COO\cdot$), follows the same reaction order as the reaction of A→P.
   a) The half-life of benzoyl peroxide at 100 °C is 19.8 min. Calculate the rate constant for the reaction. (20 pts)
   b) If the half-life of benzoyl peroxide is 7.30 h, or 438 min, at 70 °C, what is the activation energy for the decomposition of benzoyl peroxide? (20 pts)
   c) During the decomposition of benzoyl peroxide (denoted as B) in a solvent, the benzoate radicals $C_6H_5COO\cdot$ will also attack the solvent, resulting in another free radical. R· represents any free radical. R· can recombine to yield RR or react with B to give a product (denoted as X) and R·. The concentration of R· depends on the rate constants of the following three simplified elementary steps:

Step 1:	$B \rightarrow 2R\cdot$	rate constant = k_1
Step 2:	$2R\cdot \rightarrow RR$	rate constant = k_2
Step 3:	$R\cdot + B \rightarrow X + R\cdot$	rate constant = k_3

   Derive and write the expression of the concentration of free radicals (R·) in terms of the concentration of benzoyl peroxide (B) and the rate constants of the elementary steps at the steady state. (10 pts)

Useful information:

$$\int \frac{dx}{x} = \ln|x| + C$$

$$k = Ae^{-E_a/RT}$$